

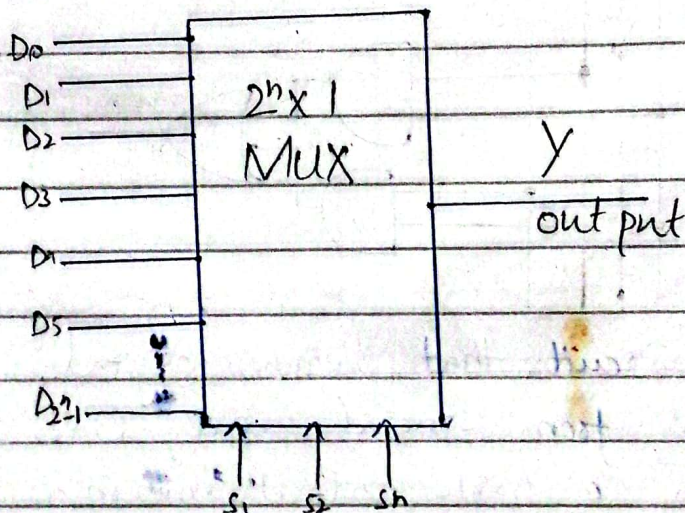
Multiplexes (MUX)

→ Multiplexes is a combinational logic circuit used to select only one input among several inputs based on selection lines.

→ This can act as digital Switch.

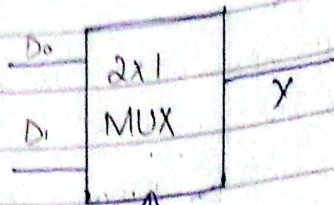
→ This is also called as data selector.

→ For a MUX there can be 2^n inputs, n selection lines & only one output based on selection lines.



Example:-

2x1 MUX Design



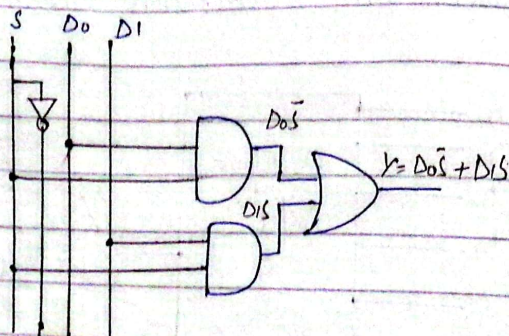
no of selection lines = $2^n = 2^1 = 2^1 = 1$

Truth table:-

S	Y
0	D ₀
1	D ₁

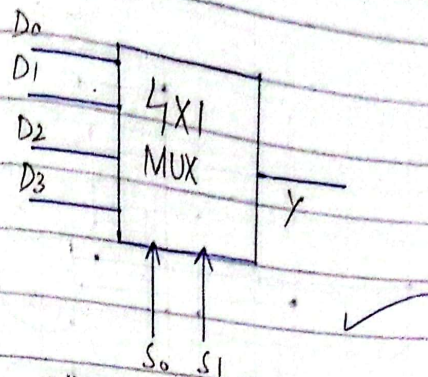
$$Y = D_0 \bar{S} + D_1 S$$

Circuit Diagram :-



A circuit that switches digital data from several input lines onto a single output line in a specified time sequence.

4x1 MUX



$2^n \times 1$
I/P O/P

$$2^n = 2^2 = 4$$

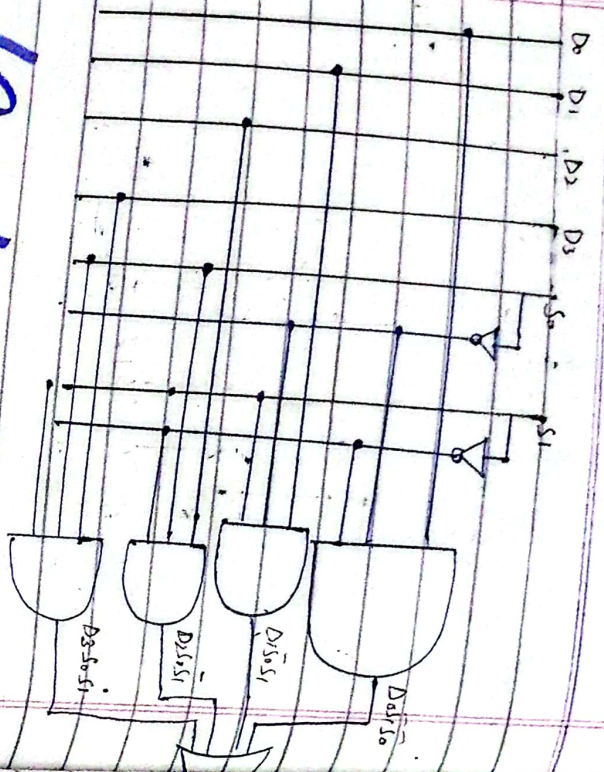
$$n=2$$

Truth table:-

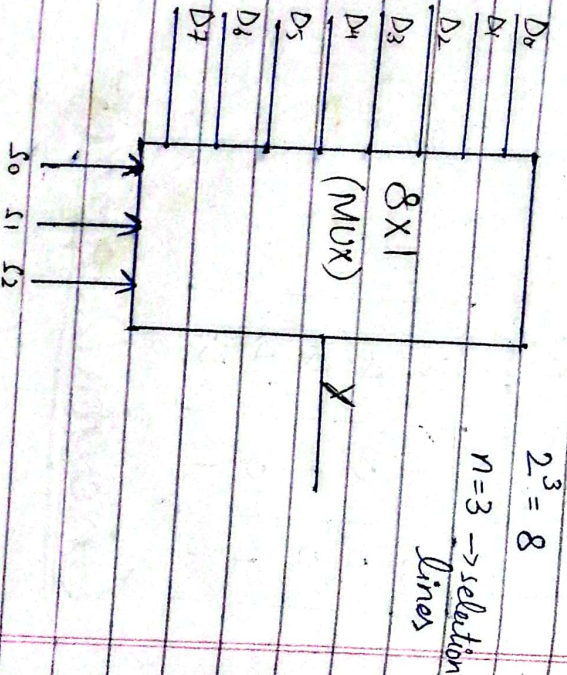
S ₀	S ₁	Y
0	0	D ₀
0	1	D ₁
1	0	D ₂
1	1	D ₃

$$Y = D_0 \bar{S}_0 \bar{S}_1 + D_1 \bar{S}_0 S_1 + D_2 S_0 \bar{S}_1 + D_3 S_0 S_1$$

Circuit Diagram :-



8x1 Mux



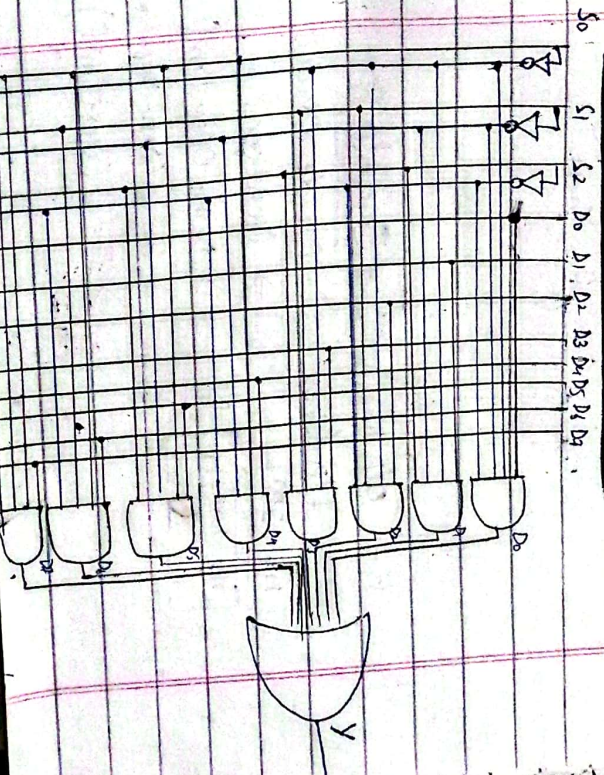
$2^3 = 8$
 $n=3 \rightarrow$ selection lines

Truth table:-

Input			Output
S ₀	S ₁	S ₂	Y
0	0	0	D ₀
0	0	1	D ₁
0	1	0	D ₂
0	1	1	D ₃
1	0	0	D ₄
1	0	1	D ₅
1	1	0	D ₆
1	1	1	D ₇

$$Y = D_0 \bar{S}_0 \bar{S}_1 \bar{S}_2 + D_1 \bar{S}_0 \bar{S}_1 S_2 + D_2 \bar{S}_0 S_1 \bar{S}_2 + D_3 \bar{S}_0 S_1 S_2 + D_4 S_0 \bar{S}_1 \bar{S}_2 + D_5 S_0 \bar{S}_1 S_2 + D_6 S_0 S_1 \bar{S}_2 + D_7 S_0 S_1 S_2$$

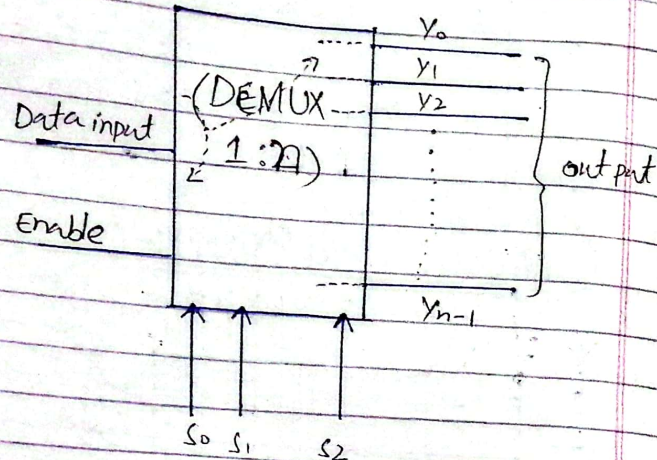
Circuit Diagram:-



vertical digital display

Demultiplexer (DMUX)

- One input and many output.
- Reverse Operation of multiplexer.
- One to many ckt or data distributor.



$n = \text{O/P lines}$
 $m = \text{I/P lines}$

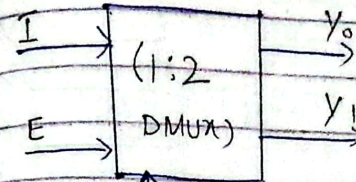
$$n = 2^m$$

$$m = \log_2 n$$

Enable

- Decide circuit is
- Operational
- Non-Operational
- $E = 0$ (Non-operational)
- $E = 1$ (Operational)

1:2 DEMUX



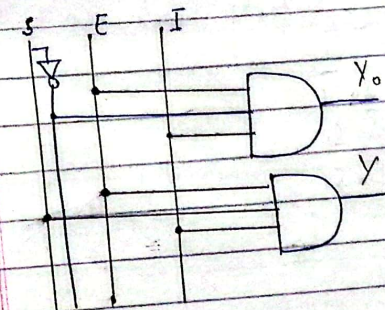
$$m = \log_2 n = 1$$

Truth table:-

E	S	Y ₀	Y ₁
0	0	0	0
0	1	0	0
1	0	1	0
1	1	0	1

$$Y_0 = \bar{E}\bar{S} \cdot 0 + \bar{E}S \cdot 0 + E\bar{S} \cdot 1 + ES \cdot 0 \Rightarrow E\bar{S}I$$

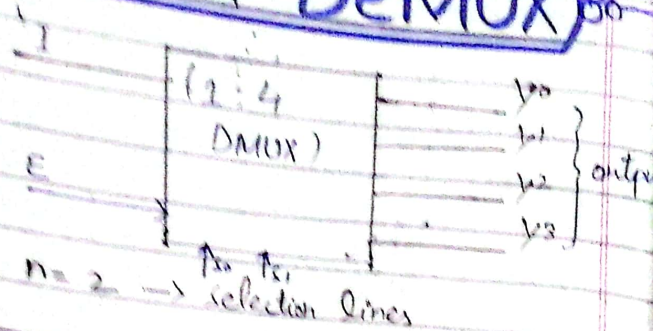
$$Y_1 = ESI$$



Applications of DMUX

- Data distribution
- Serial to parallel converter
- 1 to many circuit

1 to 4 DeMUX



S_0	S_1	E	Y_0	Y_1	Y_2	Y_3
0	0	0	0	0	0	0
1	0	0	0	0	0	0
0	1	0	0	0	0	0
1	1	0	0	0	0	0
0	0	1	1	0	0	0
1	0	1	0	1	0	0
0	1	1	0	0	1	0
1	1	1	0	0	0	1

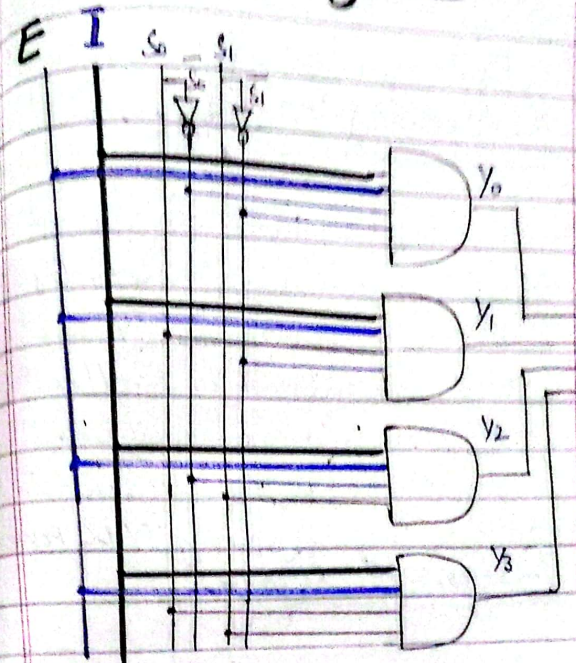
$$Y_0 = IE \bar{S}_0 \bar{S}_1$$

$$Y_1 = IE \bar{S}_0 S_1$$

$$Y_2 = IE S_0 \bar{S}_1$$

$$Y_3 = IE S_0 S_1$$

Circuit Diagram



Q# Generally, how a decoder can be used as dmux?

A decoder can be used as multiplexer by using the input lines for data selection and enable line for data input.

BCD to 7-segment

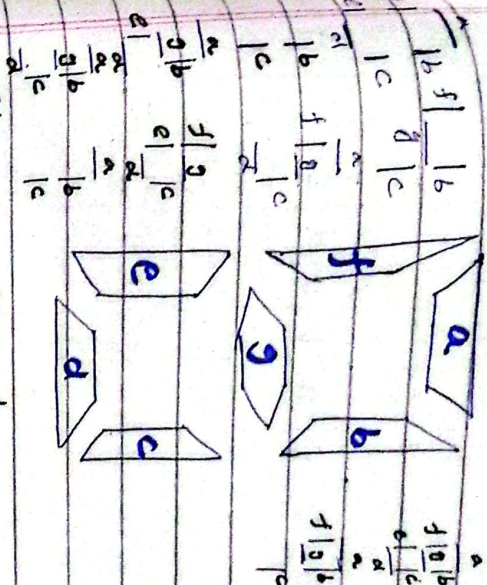
Decoder :- SSD

$\Rightarrow$ BCD to 7-segment decoder is a special decoder which can convert decimal coded decimals into another form which can be easily displayed through a 7-segment display. SSD takes generally 5V DC power.

7-segment display:

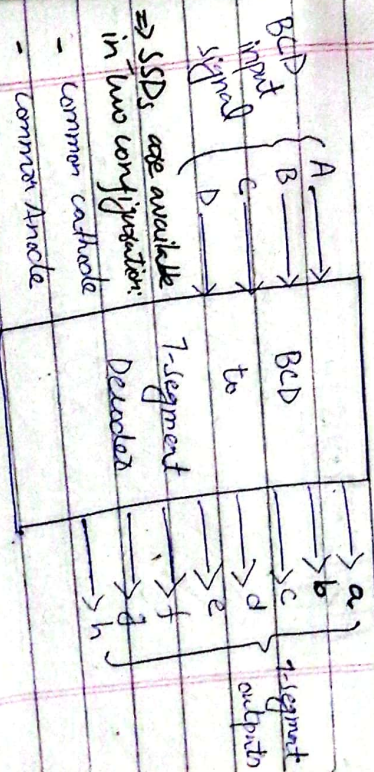
- It is a digital device that can be used for displaying decimal numbers, alphabets, and characters.
- 7-segment display contains 7-LED segments arranged in a shape.
- Generally, there are 8 input pins in a 7-segment display.

1 input pins for each of the 7 LEDs and one pin for the common terminal.



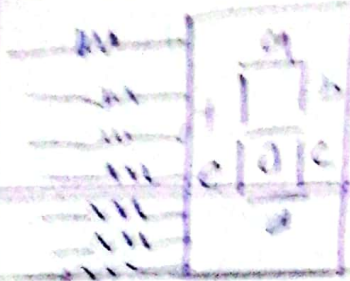
It generally has 4 input lines and 7 output lines.

Logic diagram:



$\Rightarrow$ SSDs are available in two configurations:

- common cathode
- common Anode



7-segment LED Display

Truth table - for common cathode type BCD to seven segment decoder.

D	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

Applications:-

=> Seven-segment displays are used to display the digits in calculators, clocks, various measuring instruments, digital watches and digital counters.

K-maps:

for a:

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$$F(ABCD) = \bar{B}\bar{D} + C + \bar{B}D + A$$

for b

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	X	X	X	X
10	1	1	X	X

$$F(ABCD) = \bar{B} + \bar{C}\bar{D} + CD$$

for c

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	X	X	X	X
10	1	1	X	X

$$F(ABCD) = \bar{C} + D + B$$

for d

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	X	X	X	X
10	1	1	X	X

$$F(ABCD) = \bar{B}\bar{D} + \bar{B}C + B\bar{C}D + \bar{C} + A$$

for e

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	X	X	X	X
10	1	0	X	X

$$F(ABCD) = \bar{B}\bar{D} + \bar{C}\bar{D}$$

for f

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$$F(ABCD) = \bar{C}\bar{D} + \bar{B}\bar{C} + \bar{B}\bar{D} + A$$

for g

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$$F(ABCD) = \bar{B}C + \bar{B}\bar{C} + A + \bar{B}\bar{D}$$

Q# what is difference b/w multiplexers and demultiplexers?

Multiplexers

- It is known as data selector.
- Multiplexer is a digital switch.
- It has many input one output. It means many to one.

Applications:

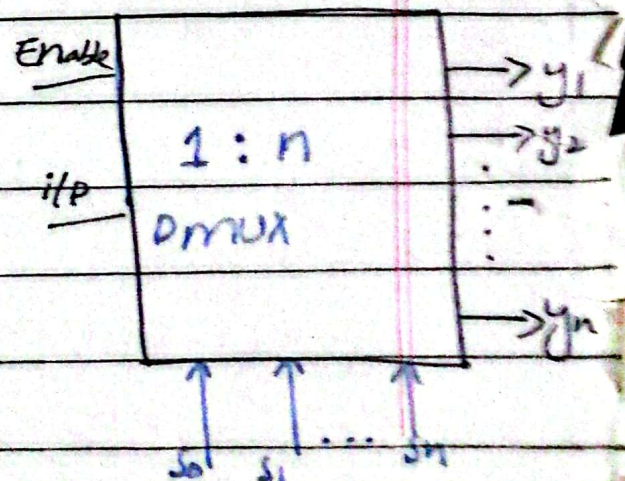
- Computer memory
- Telephone network
- Handle two types of data analog and digital. for analog application mux are build of delays and transistors switches.
- For digital application they are built from standard logic gates

Demultiplexers

- It is known as data distributor
- DeMUX is a digital circuit.
- It has one i/p and many output. It means one to many.

Applications:

- used for reconstruction of parallel data and ALU circuits.
- Communication system
- Serial to parallel Converter.



Date _____

Day MTWTFS

Q# What is difference between Demultiplexers and decoders?

Demultiplexers

Def:- A dmux is a circuit which takes only one input and switches it to one of the several outputs with the help of selection line.

Number of inputs → A DMUX accepts the data as input.
→ only one
Data input

Number of o/p The output depends on the number of selection lines. 2^n .
Need of selection lines Required

Main function Used for Switching.

Decoder

A decoder is the circuit which decodes the input signal fed to it with the help of control signal.

A decoder accepts control signal as input.
→ one or more signals can be given as input.

The output depends on the number of control signals 2^n .
Not required, but control signals and enable signal is required.

Used for decoding of the signal encoded input terminal.